RESEARCH · 03

The True *Contributors*.

WHO ACTUALLY MOVES THE TABLE? · A THREE-LAYER EXPECTED-POINTS ATTRIBUTION

2026·06

renball

TL;DR: three sentences

- We take project 02's time-aware expected-points (EP) value function and credit a THIRD touchpoint per goal (the pre-assist, SCA2) alongside the scorer and the assister.

- Across 39,381 goals in six leagues and seven seasons, the per-90 board still leads with the consensus elite (Lionel Messi, 1.14 EP/90), the framework clears the eye test before it surfaces surprises.

- The marquee finding: Julian Brandt collects 40.93 EP from pre-assists alone, second in Europe only to Messi (51.69), and 39% of his total value comes from that single layer.

THE QUESTION

Three touchpoints, *two* counted

Every goal is built by a chain of contributions. The scoreboard credits the end of the chain and forgets the rest.

SCORER	Puts the ball in the net. Always credited: the goal goes on his record.
ASSIST · SCA1	The final pass. Sometimes credited, when the box score bothers to log it.
PRE-ASSIST · SCA2	The pass before the pass. Almost never credited: it disappears into the squad page.

→ *Credit all three. Then ask who the model elevates that the box score does not.*

DATA & METHOD

What the framework runs on

39,381	6	7	6,516
GOAL EVENTS	LEAGUES	SEASONS	PLAYERS W/ META

```
three_layer credit  =  scorer + SCA1 + SCA2     (each role = 1 credit · EP summed)
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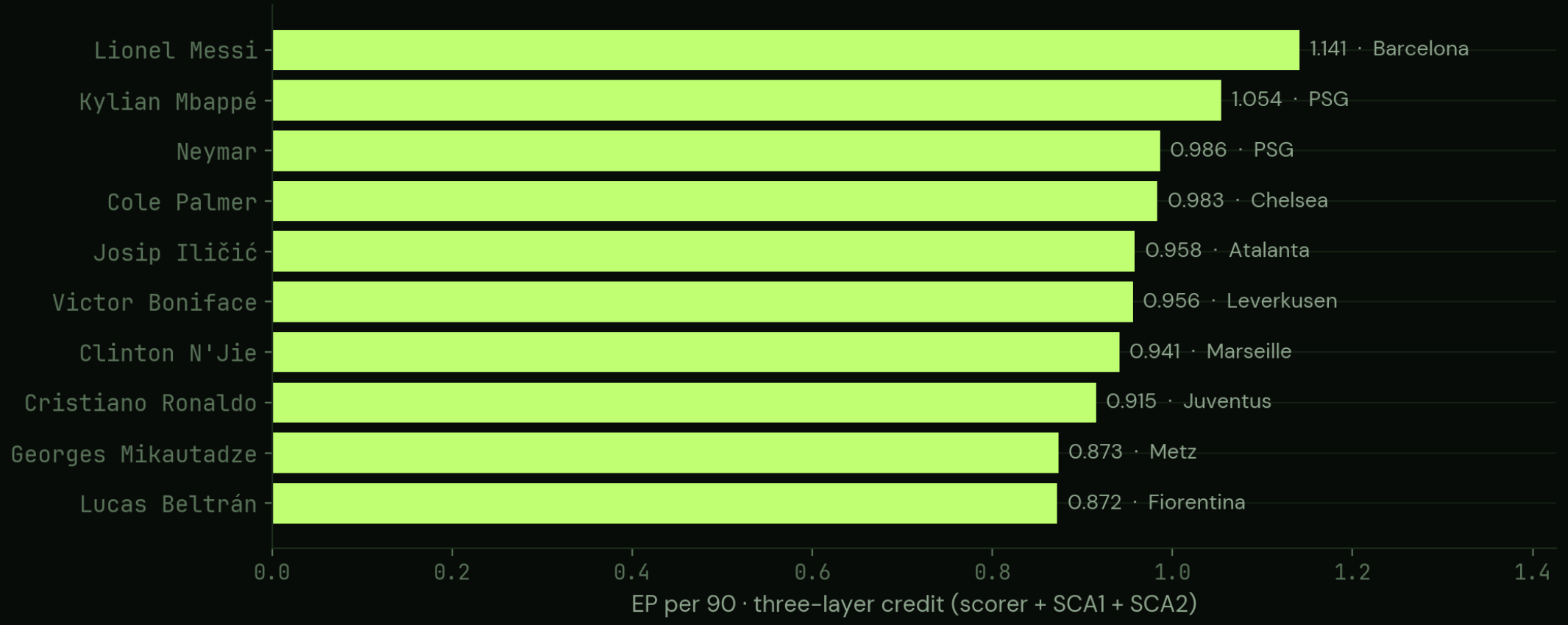
Inherits project 02's value function verbatim: a goal is priced by the swing in expected league points it produces, conditioned on game-state and minute.

Per-90 = (EP / minutes) × 90. Minutes come from FBref summary tables, Top-5 leagues only, so every per-90 board excludes Eredivisie.

Absolute-EP boards keep all six leagues; per-90 boards are Top-5 only.

Q1A · FULL-CREDIT LEADERBOARD

The model clears the *eye test*



Top 10 by EP/90, ≥ 1,000 minutes, Top-5 leagues. The consensus elite leads, which is the point. Recover what everyone knows before surfacing surprises.

Does *weighting* the credits change the answer?

- Re-run the same board with credits weighted 0.40 / 0.35 / 0.25 (scorer / SCA1 / SCA2). EP only; the minutes threshold is unchanged.
- The top eight is essentially identical. The biggest movers are pure scorers gaining a little under the heavier scoring weight: Lewandowski +6, Dovbyk +7.
- Creator-heavy names slip a few places: Jallow −9, Ljajić −9.
- That the tuned weights reproduce the simple 1/1/1 board IS the result: the ranking is not an artefact of the credit split.

Full rank-shift table on the dashboard · Q1 Rank Shift (A vs B)

The model's MVPs, *season by season*

LA LIGA · 2022-23

Nicolas Jackson

Villarreal · 1.211 EP/90 over 1,598 minutes (26 credits).

La Liga's top per-90 pick the season before he joined Chelsea.

5 LEAGUES COMBINED · 2020-21

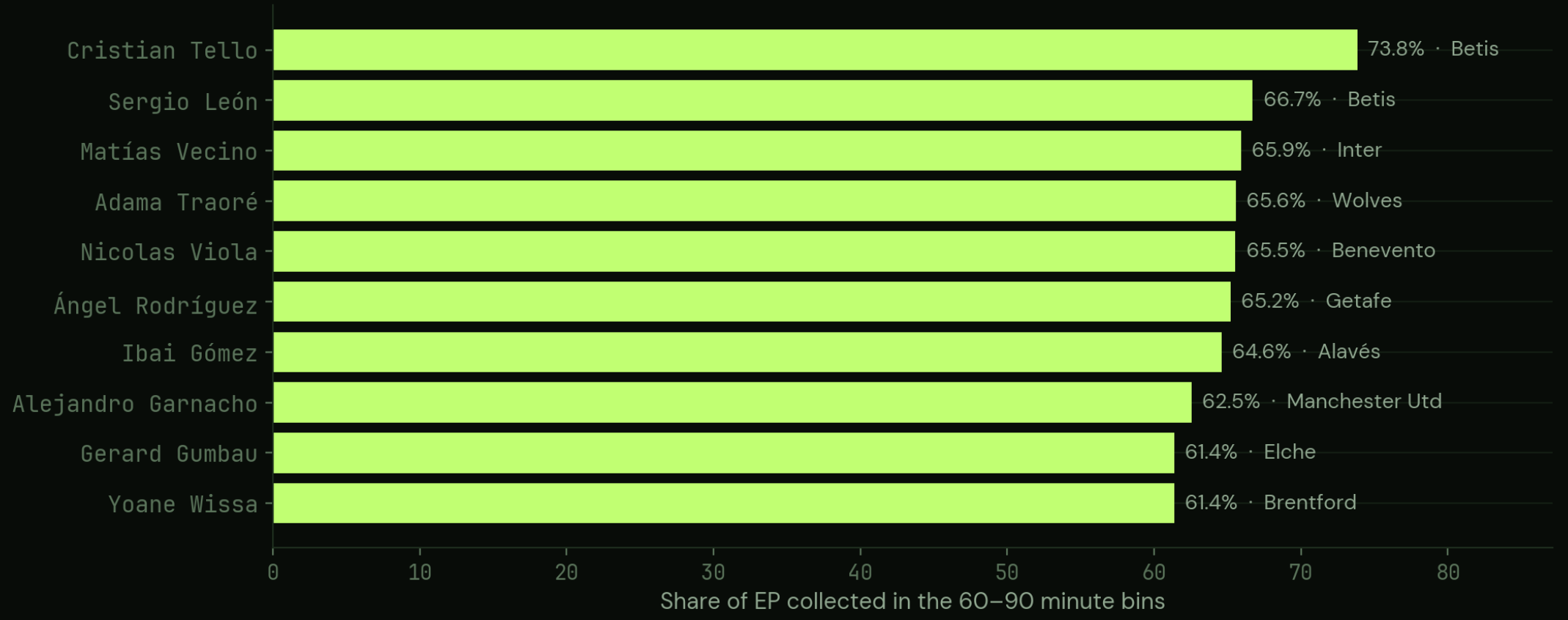
Luis Muriel

Atalanta · 1.605 EP/90 (43 credits).

The model's Ballon d'Or pick that season, not what the trophy cabinet recorded.

Model's top EP/90 pick per (league × season) vs the real award winner, matched in 11 of 40 cells (28%). The model prices leverage, not reputation. Top-5 only.

The late-game *substantives*



Top 10 by late share, ≥ 30 three-layer credits (super-subs filtered out). Includes all six leagues. Tello takes 73.8% of his value after the 60th minute.

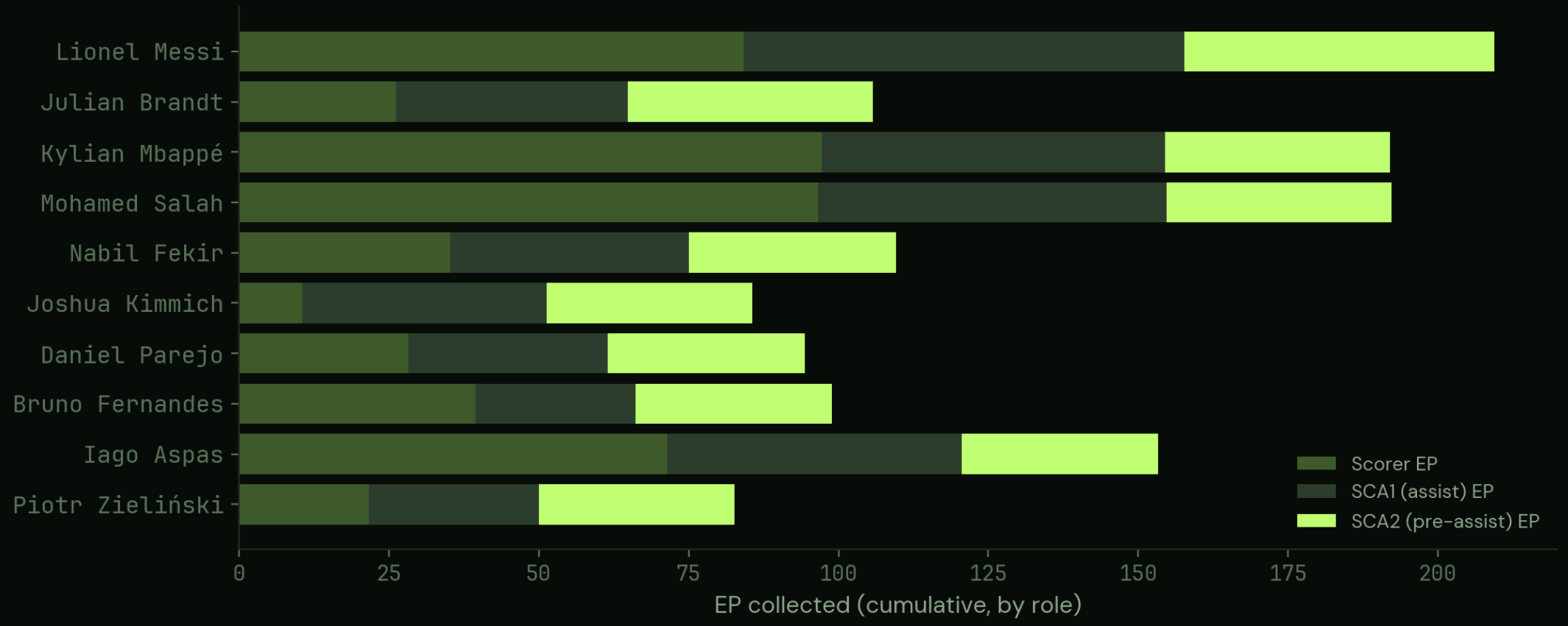
Position-adjusted · *outfield* only

POSITION	#1 PICK	EP/90	MINUTES	CREDITS
Defender	Samuel Iling-Junior	0.785	1,128	11
Midfielder	Adem Ljajić	0.845	2,185	26
Forward	Lionel Messi	1.141	16,516	356

Top EP/90 per modal position, ≥ 1,000 minutes. GK excluded: a structural blind spot of the framework (top GK ~0.1 EP/90 vs ~1.1 for forwards). De Bruyne is the most-sampled elite pick anywhere here: 14,674 minutes, 206 credits.

Q6 · MARQUEE FINDING

The Hidden *Architects*



Top 10 SCA2 EP collectors, stacked by contribution role. Bright accent = the pre-assist layer. One player is structurally different from everyone else.

SPOTLIGHT

What the model *exists to find*

PRE-ASSIST ARCHITECT

Julian Brandt

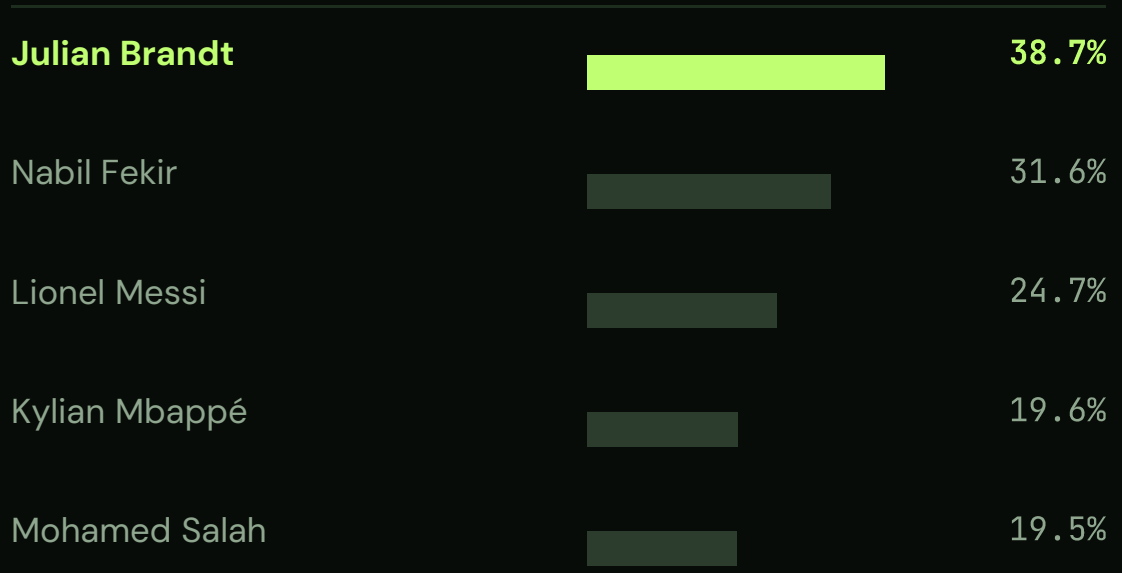
Borussia Dortmund · MF

40.93

EP COLLECTED FROM SCA2 ALONE

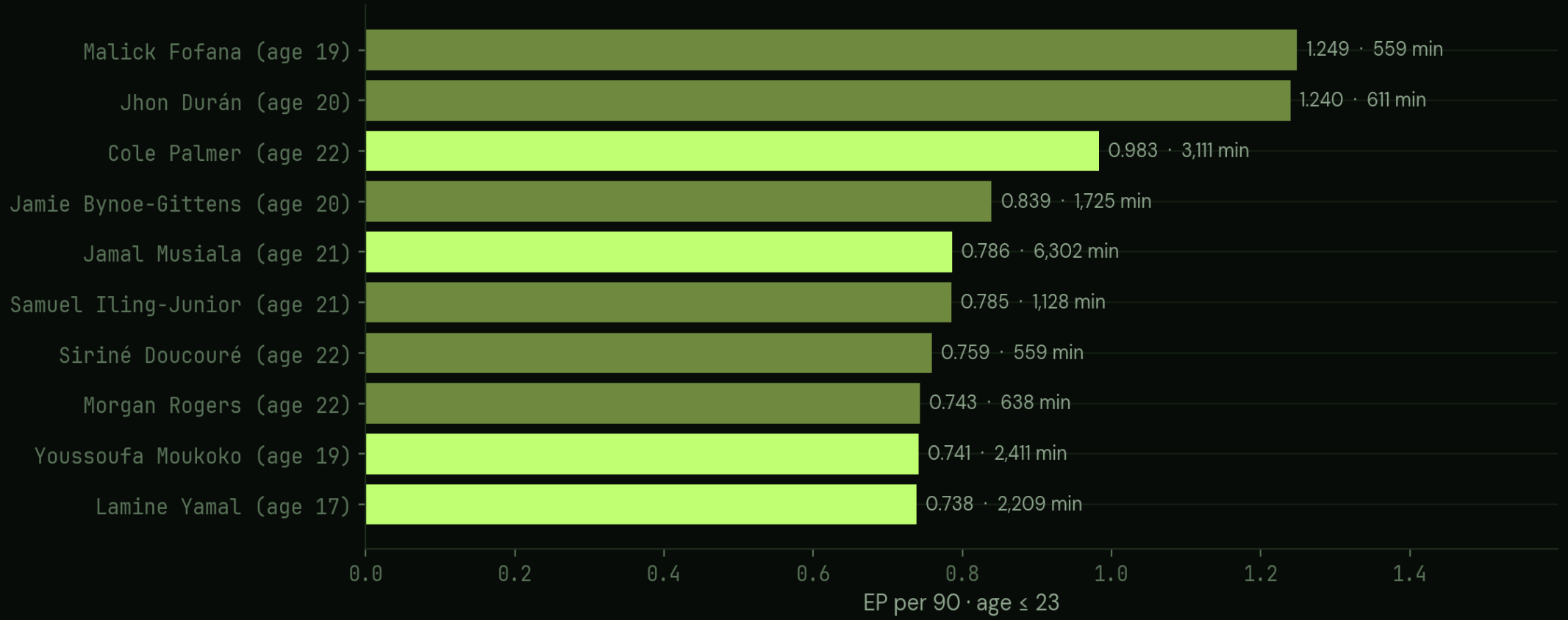
Second in Europe only to Messi (51.69). The pre-assist is his single largest contribution layer.

SCA2 SHARE OF TOTAL EP



Brandt's profile (25% scorer / 37% SCA1 / 39% SCA2) is shared by no pure goalscorer.

Young players to *watch*



Top 10 EP/90, age ≤ 23, ≥ 500 minutes. Dim bars = under 2,000 minutes (low sample). Yamal sits #10 on 2,209 minutes at age 16.9, real volume, not a cameo.

IMPLICATIONS

What this *changes*

- Contribution is a chain, not a tally. Pricing the pre-assist surfaces value the goal-and-assist box score structurally cannot see.
- Recruitment: a player like Brandt, whose largest layer is the pass before the pass, is materially mis-valued by conventional output metrics.
- The ranking is stable. Re-weighting the three layers barely moves the board, so the discoveries beneath the elite are not an artefact of the credit split.
- The dashboard exposes all seven boards with per-league and per-season drill-down, so any analyst can apply their own thresholds.

LIMITATIONS

What this *isn't*

- Not xG-weighted; penalties not separated; own goals missing from the shot data (inherited from project 02).
- Credit caps at SCA2: build-up beyond the pre-assist isn't captured.
- Per-90 boards exclude Eredivisie: the minutes source covers the top five leagues only.
- No opinion on goalkeepers: the three credited touchpoints are ones GKs rarely occupy.
- Observational, not causal: EP is the gap between states a goal connects, not its counterfactual effect.

Full discussion: methodology.html · §5

Every player, every league, every season.

Full methodology, source code and data:

renball.com

/PROJECTS/03-TRUE-CONTRIBUTORS/

→ analysis.html	editorial write-up · Chart.js + custom SVG visuals
→ dashboard.html	interactive Tabulator dashboard · 7 datasets · drill-down
→ methodology.html	three-layer attribution · seven questions · nine limitations
→ code/build.py	Python pipeline: project 02 framework + SCA2 enrichment